

2025_EL4_KO_data

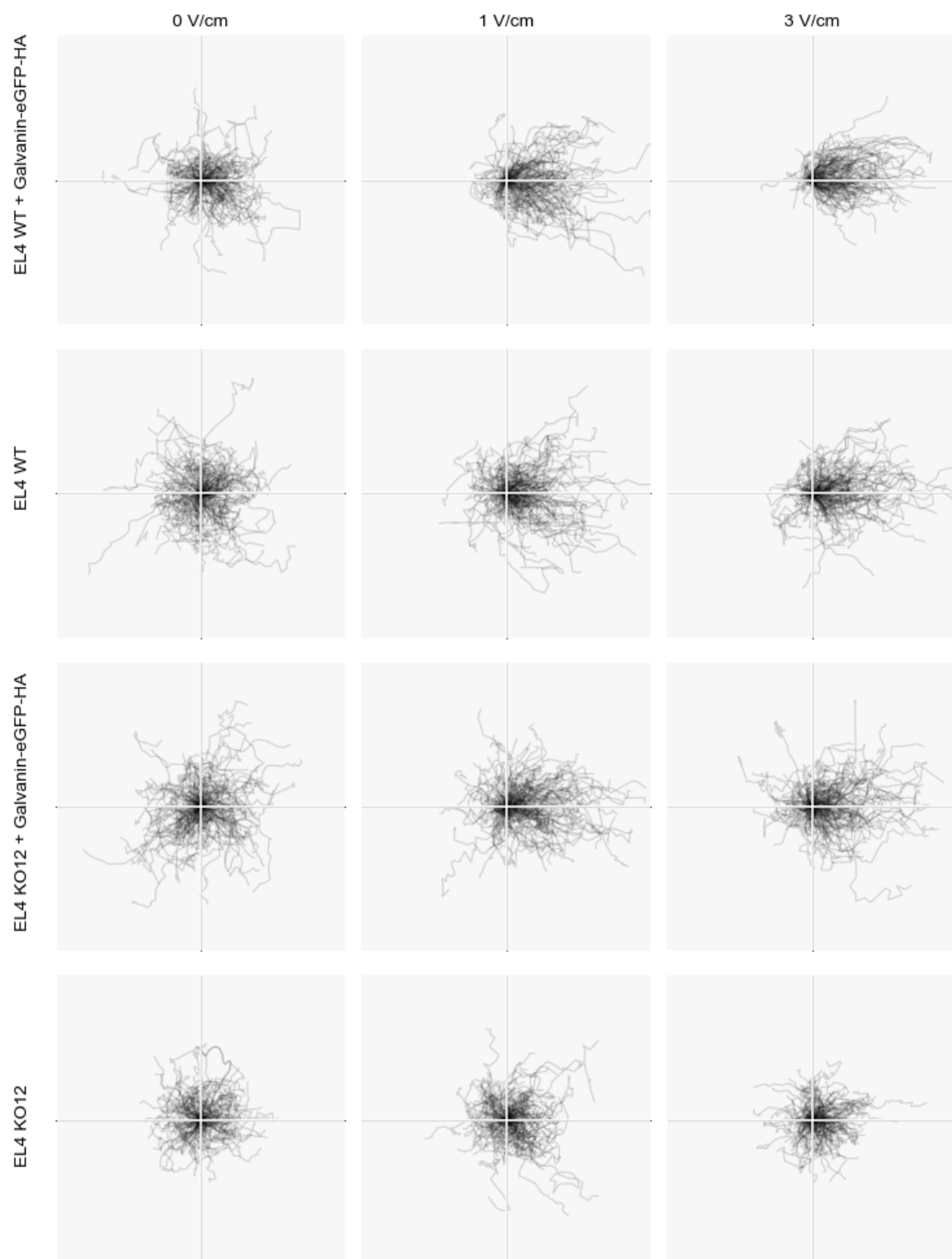
June 8, 2025

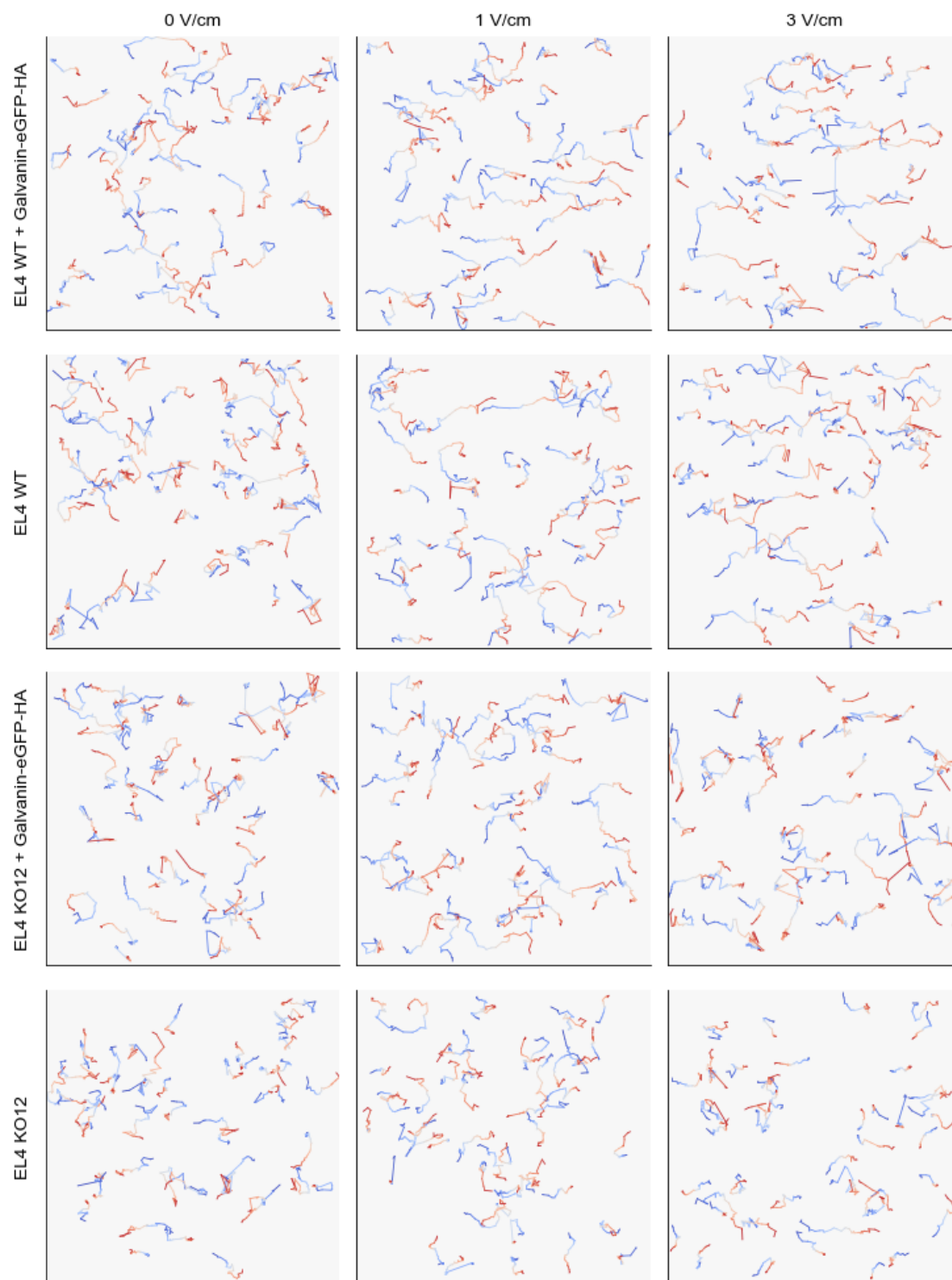
```
[22]: array(['EL4 K012', 'EL4 K012 + Galvanin-eGFP-HA', 'EL4 WT',  
          'EL4 WT + Galvanin-eGFP-HA'], dtype=object)
```

1 Plotting of EL4 migration data

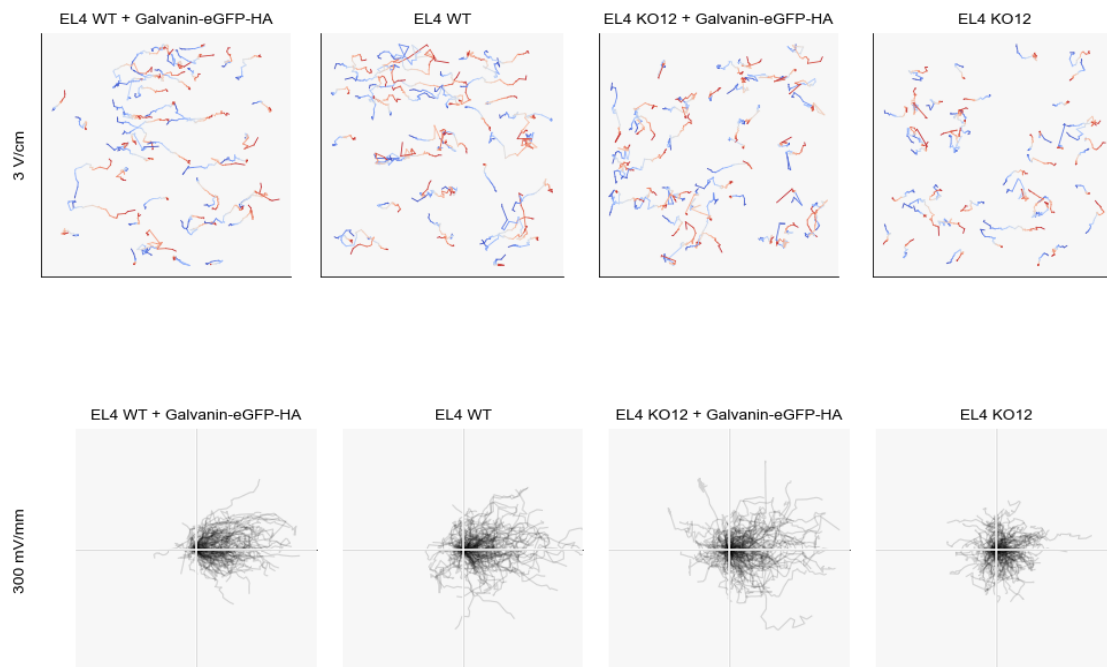
- I've included data from all four cell lines (wild-type, wild-type + Galvanin-GFP, Galvanin Knockout, and Galvanin Knockout + Galvanin-GFP).

1.1 Tracks

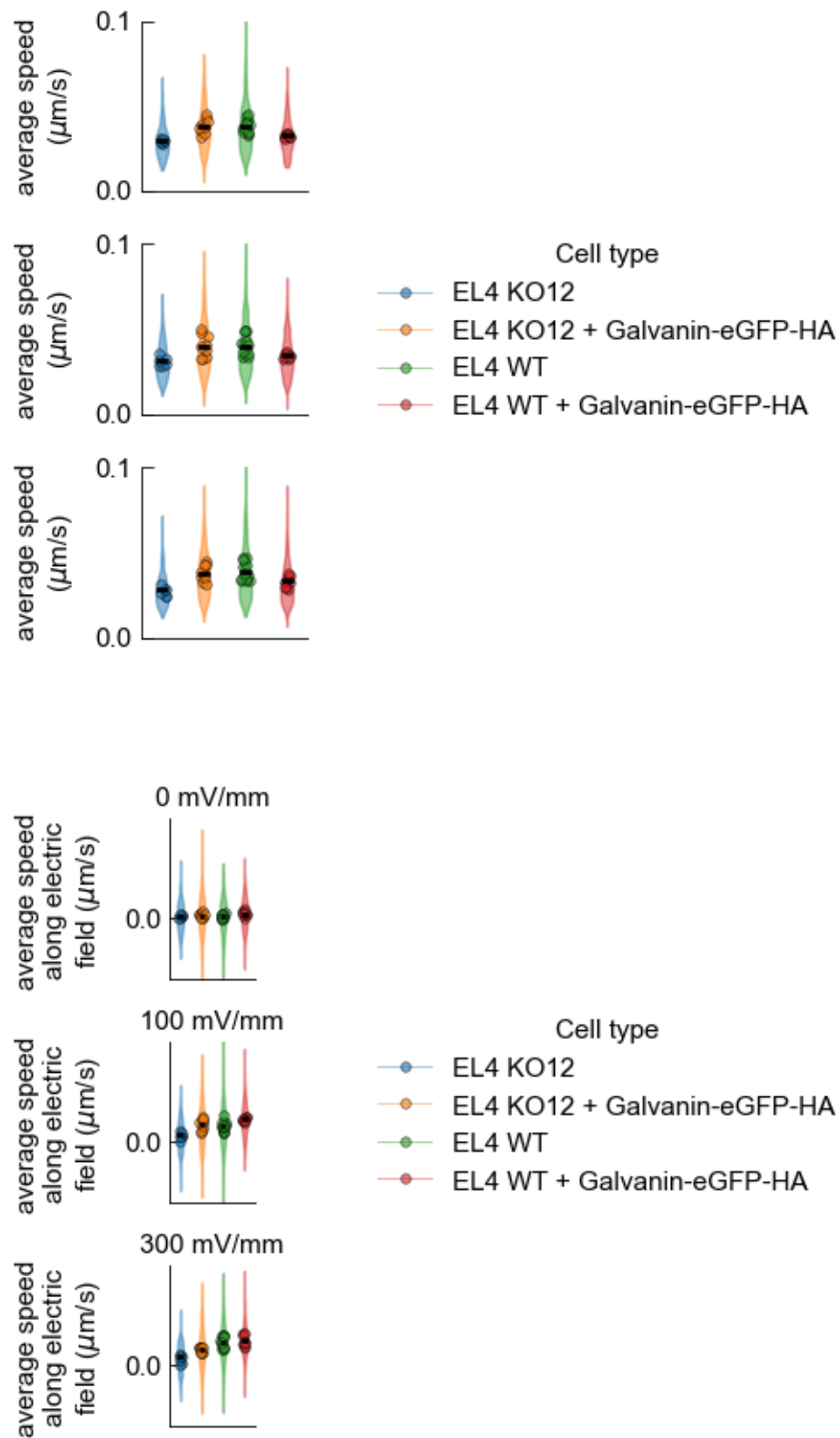




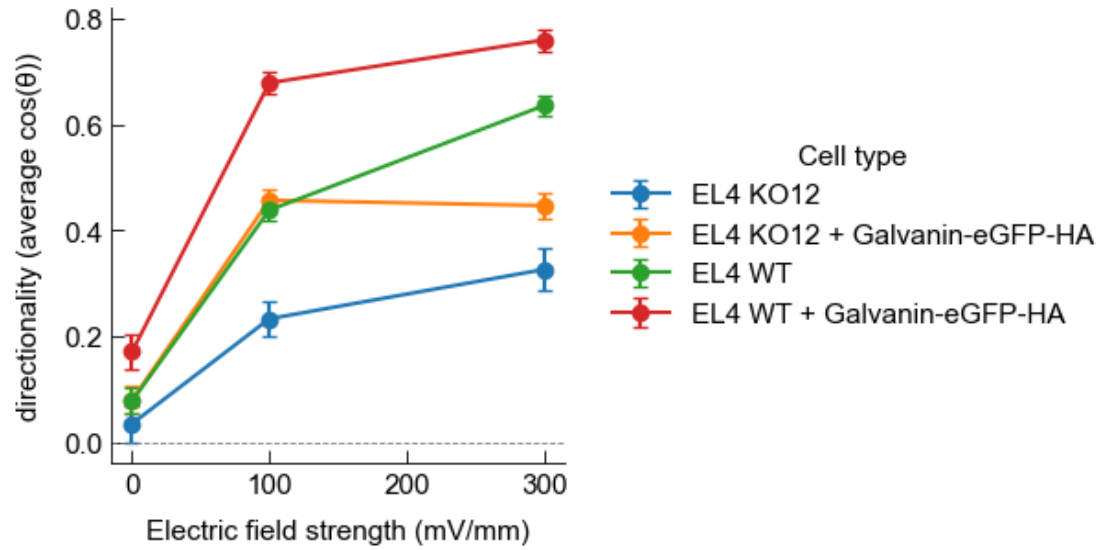
1.2 Here are only plots of 300 mV/mm condition only - probably more useful for main text figures.



2 Speed related

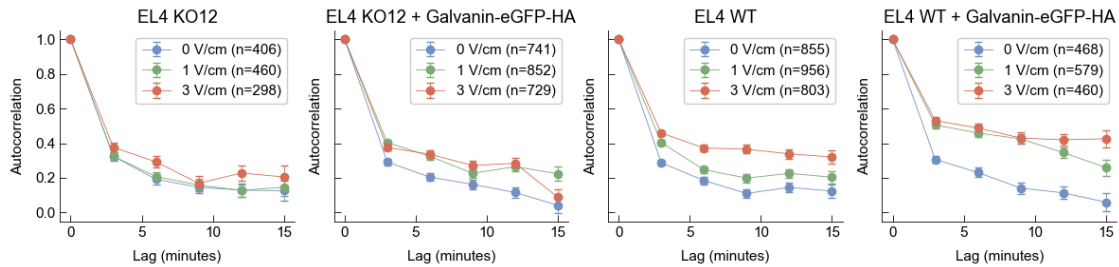


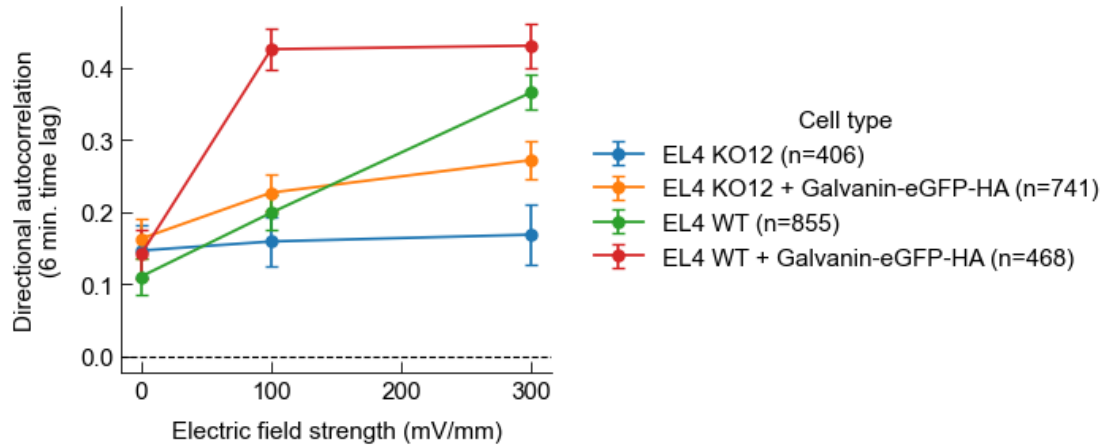
3 Directionality (average cosine theta)



4 Autocorrelation Analysis

This uses cosine theta between trajectory vector and electric field (the regular approach we've been taking).





4.1 Autocorrelation, but relative to 0 mV/mm

Here we plot relative to the 0 mV/mm condition to try to account for residual persistence in the absence of an electric field. (i.e. subtraction of autocorrelation value from the 0 mV/mm condition).

